Advanced Optimiser Tab Manual

Overview

The **Optimiser Tab** determines the best combination of **Spot Size**, **Laser Repetition Rate**, and **Scan Speed** (or Dwell Times) to achieve your analytical goals. It works by "working backwards" from your desired signal quality (SNR) and hardware limits to find the most efficient sampling capability.

User Interface

1. Left Panel: Settings & Configuration

Top Controls

- **Import Optimisation File**: Loads or refreshes the optimization input data from a file in your iolite session.
- **Settings**: Opens the detailed **Hardware Configuration** dialog.

Hardware Configuration Dialog

Accessible via the "Settings" button. This establishes the physical limits of your system. A version label (e.g. `Version: dev`) is displayed at the bottom-left of this dialog, alongside **GitHub Repo**, **Report Issue**, and **Check for Updates** buttons to interact with the repository on GitHub.

- **ICP-MS Hardware**:
- **Manufacturer/Model**: Presets minimum dwell times and switching speeds.
- **Custom Models**: Choose "Custom Model" (ICP) or "Custom Laser" (Platform) to manually define limits:
    - **ICP Custom Fields**:
    - **Type**: Quadrupole, Sector-Field, Multi-Collector (MC), or TOF.
    - **Allowed Dwell Times**: (MC only) Comma-separated list of valid integration times.
    - **Min Dwell / Dwell Resolution**: (Quad/TOF) The hardware speed and rounding limits.
    - **Washout Margin (%)**: (TOF only) A percentage "Signal Capture Buffer". Extends the target washout time proportionally to ensure the entire Single Pulse Response is captured natively within the stage's speed budget.
    - **Laser Custom Fields**:
    - **Mode**: Choose "Maximum Rep-Rate" (continuous) or "Discrete" (specific allowed frequencies).
    - **Max Speed**: Physical limit of the sample stage.
    - **Rep-Rate Resolution**: Custom rounding resolution for the laser repetition rate.
- **Laser Hardware**:
- **Platform/Source**: Presets maximum repetition rates and frequency precision.
- **Cell Type**: Defines the maximum stage speed (washout behavior is handled separately).

Input Parameters

Basic setup for your experiment.

- **Initial Spot Size**: The starting point for optimization (often adjusted automatically).
- **Washout**: The time (ms) required for the signal to decay (usually determined via the **SPR Tab**). Supports extremely fast sub-1ms definitions (e.g., `0.25 ms`).
- **Initial Rep-Rate**: A baseline frequency guess (Hz).
- **Number of Analytes**: Specifies how many isotopes/analytes are measured (saved persistently).
- **Suggest PreScan Params**: Triggers a calculation dialog presenting optimal settings for a PreScan scan:
- **Dwell Time per Analyte**: `washout_ms / num_analytes`, bounded by minimum dwell and rounded to the instrument's dwell resolution.
- **Target Cycle Time**: `dwell_time * num_analytes`.
- **Suggested Rep-Rate**: Dynamically swept starting from the oversampling limit ($2.0/\text{washout s}$) to select the lowest rate that achieves a statistical Lockwood RSD $\leq 5\%$.
- **Suggested Stage Speed**: `spot_size / cycle_time`.
- **Suggested Dosage**: `rep_rate * cycle_time` (pulses/pixel).
- **Suggested Overlap**: Spatial overlap distance and percentage between adjacent pulses.
- **Mode**:
- **Spot**: Analysis of a single location.
  - **Total Pulses**: Total laser pulses per spot.
- **Line**: Continuous scanning along a path.
  - **Line Length**: Total distance (μm).
- **Imaging**: 2D mapping (raster scan).
  - **Image Width / Height**: Dimensions of the area (μm).

Optimisation Parameters

The "Goals" you want the algorithm to achieve.

- **Sync Strategy**:
- **Adaptive Integer Sync (Auto)**: Automatically sweeps rep-rates and acquisition times to guarantee a perfect integer pulse count sync (0% rounding error) close to your target.
- **Strict Pulse Target (Manual)**: Snaps directly to the exact target pulses (may lead to non-integer pulses due to hardware rounding).
- **Oversampling (Time-Driven)**: Purely time-driven target based on the Lockwood oversampling RSD target, ignoring snapping.
- **Combined Integer Sync (Auto)**: Combines oversampling with snapping by sweeping frequencies above the oversampling minimum to find a perfect integer count.
- **Sync Target (Pulses)**: The target number of laser shots that must occur during a single mass spectrometer integration (`Acq Time`) to ensure signal stability. Used in integer sync modes.
- **Sync Target (RSD %)**: The target Relative Standard Deviation (RSD) limit for signal stability during steady-state oversampling modes (usually set to `5.0%`).

- **Prefer Exact Pulses per Acq**: (For Auto sync strategies only). When checked, forces the auto-sync search algorithm to prioritize finding a laser rep-rate that delivers *exactly* the requested integer number of pulses per cycle, rather than settling for the closest available integer sync.
- **Target SNR (Sigma)**: The desired Signal-to-Noise ratio for the limiting isotope.
- **Robust Methodology**: The optimiser calculates a **Detection Ratio** based on the scientific Critical Level (L_c), then scales it back to a familiar "Sigma SNR" for the UI. This provides a more rigorous assessment than simple background standard deviation:
    1. **Noise Floor**: Determined by the maximum of the theoretical **Poisson Variance** (counting statistics) and the **Observed Variance** (experimental jitter) of the background.
    2. **Critical Level (L_c)**: Calculated using the **Square Root Transform** rule (Stapleton's Rule) for variance stabilization. This defines the 95% confidence threshold for detection.
    3. **Sigma-Equivalent Scaling**: The final Detection Ratio is multiplied by $z\sqrt{2}$ (≈ 2.326) so that a value of **10.0** in the UI corresponds to the standard 10-sigma **Limit of Quantification (LoQ)**.
- This approach is conservative and robust, ensuring that targets represent accurate confidence intervals rather than just multiples of potentially "lucky" low-noise baselines.
- **Min Duty Cycle**: Sets a floor for efficiency (Time Measuring / Total Time).
- If the calculated duty cycle is too low (mostly settling time), the optimiser will increase dwell times to improve the ratio.
- **Minimum SNR**: Lower limit for acceptance; channels below this may trigger warnings.
- **Scale Signal with Rep-Rate**:
- **Checked (Linear Scaling)**: Assumes signal intensity (CPS) increases linearly with Frequency (Hz).
    - **Why?** Higher Rep-Rate delivers more pulses per second, ablating more material per second, which generates a higher ion signal.
    - **Use Case**: **Geological Samples** (e.g., blocks/mounts) where there is "infinite" depth. As the laser fires active shots, it drills deeper into fresh material, maintaining the signal increase.
- **Unchecked (Fixed Sensitivity)**: Assumes signal intensity is determined by Spot Size alone, regardless of frequency.
    - **Use Case**: **Thin Sections / Bio-Imaging**. If the laser already fully ablates the entire thickness of the sample (e.g., a 30μm tissue section) with a single shot, firing faster *does not* generate more signal—it simply consumes the sample faster. Checking this box here would dangerously overestimate your SNR.
- **Avoid Gaps**:
- **Goal**: Prevents "striping" in images by ensuring the stage doesn't move faster than the laser can fire.
- **How**: Constraints the calculation so `Speed <= Spot Size * Rep Rate`.
- **Pros**: Guarantees full surface coverage.
- **Cons**: May force a slower stage speed or higher Rep-Rate (potentially increasing washout issues).

Optimised Settings (Results Summary)

The calculated "Ideal" instrument settings.

- **Spot Size**: The recommended beam diameter.
- **Rep-Rate**: The optimal laser frequency (Hz).
- **Speed**: The optimal stage translation speed (μm/s).
- **Acq Time / Budget**: The total integration cycle time calculated to match the laser synchronization.

- **Est. Time**: Total predicted duration of the analysis (HH:MM:SS).
- **Est. Data Points**: (Displayed in Spot/Line modes only) The predicted total number of mass spectrometer measurements that will be collected over the run.
- **Pixels Per Sec**: (Displayed in Imaging mode only) The scanning rate of pixels per second based on the optimised scan speed and spot size.

2. Right Panel: Visualisation & Details

Signal Plot (Top)

- Displays real-time signal traces for the selected channels.
- **Regions**:
- **Background (Blue)**: Time range for baseline calculation.
- **Signal (Red)**: Time range for signal intensity calculation.
- **Auto Select Regions**: Automatically finds the rising/falling edges of the signal.

Plot Controls

To maximize vertical display efficiency, controls are split into two stacked rows:

- **Row 1**: Theme override dropdown (Auto/Dark/Light), Normalise checkbox, Pan/Zoom Y checkbox, Auto-Rescale Y checkbox, Show Background checkbox.

- **Row 2**: Auto Select Regions checkbox, Background time spinboxes, Signal time spinboxes.

- **Theme**: Toggle Dark/Light mode.

- **Normalise**: Scales all traces to 0-1 range for easy comparison.
- **Auto-Rescale Y**:
- **Checked (Default/Ephemeral)**: The plot automatically snaps to the full data range. This is reset to ON whenever new data is loaded or the application starts.
- **Unchecked**: The plot keeps your current zoom level. Switch this off to "lock" your view while adjusting parameters.
- **Pan/Zoom Y**: Enables vertical zooming interaction.
- **Double-Click**: Resets the plot to show the full data range (both X and Y).
- **Interactive Region Adjustment**:
- **Shift + Click & Drag**: Hold the **Shift** key and click near a region edge (Blue or Red line) to drag it to a new location.
- **Auto-Swap Validation**: If you drag a region "start" line past its "end" line, the software automatically sorts the values on release. This ensures the start is always less than the end, preventing calculation errors.
- **Cursor Feedback**: When holding Shift, the cursor will change to a horizontal resize cursor ↔ when you are within the 1% selection "hitbox" of a region edge.

Synchronization Advisor

Displays real-time diagnostic indicators based on the active Sync Strategy:

- 🟢 **Synchronised**: Green indicator representing perfect snapping to integer pulse counts.
- 🟢 **Oversampling Mode**: Green indicator indicating no integer snapping is required (pure time-driven).
- 🟢 **Stable Steady State**: Green indicator indicating all channels are within the target RSD limits.
- 🔴 **Unstable Steady State**: Red indicator informing the user that the rep rate is too low for oversampling ($< 2.0/\text{washout s}$).
- 🟡 **Unstable Steady State**: Yellow indicator showing that the statistical Lockwood RSD is above the target limit, and listing the required dwell time adjustment to achieve stability.
- 🟡 **Unsynchronised**: Yellow indicator warning of potential beating artifacts if a non-integer pulse count is detected during manual overrides.

Optimised Dwell Time Distribution (Table)

Detailed breakdown per channel.

- **Mode**:
- **Auto**: Software calculates the optimal dwell.
- **Even**: Software splits the entire available dwell budget perfectly among all `Even` channels. Calculates exact hardware intervals so no time is lost to drift or rounding errors.
- **Exclude**: Channel is ignored.
- **Set to Min**: Forces the channel to the hardware minimum dwell.
- **Custom**: Allows manual entry of a specific dwell time.
- **Optimised Dwell Times**: The calculated integration time for each isotope.
- **SNR**: Comparison of Initial vs. Resultant Signal-to-Noise Ratios.

CSV File Importing & Preferences

When loading external standard or tuning data via a CSV file, iolite requires the timestamp column format to exactly match the format configuration selected in preferences. If a mismatch occurs, the import will fail to parse and show a dialog:

- **Error popup**: Displays detailed failure information extracted from the iolite application log.
- **Smart Suggestions**: Analyses the failed timestamp structure and suggests the exact matching preferences layout to configure in your iolite settings (e.g. suggesting `yyyy MM dd hh mm ss` or `dd MM yyyy hh mm ss`).

Workflow Guide

Step 1: Hardware Setup

1. Click **Settings**.
2. Select your **ICP-MS** and **Laser** models.
3. If your specific model isn't listed, choose "Custom" and enter the **Min Dwell Time** (ICP) and **Max Rep Rate** (Laser) manually.

Step 2: Load Data

1. Ensure your iolite session has representative data (e.g., a "Tuning" line).
2. Click **Import Optimisation File** and select your data file.
3. **Missing Dwell Times?**
   - If your data lacks embedded dwell time metadata (common with some CSV imports), a dialog will appear.
   - **Set to Global**: Apply a single dwell time (e.g., 10ms) to all channels.
   - **Individual**: Manually enter the dwell time for specific channels if they differ.
4. Verify the plot shows the signal traces.
5. **Check Regions**: Ensure the **Background (Blue)** and **Signal (Red)** regions are correctly identifying the baseline and signal plateau.
   - **Manual Selection**: Hold **Shift** and drag the lines in the plot for the fastest adjustment.
   - **Precise Selection**: Use the spinboxes in the left panel.
   - **Auto Select**: Click "Auto Select Regions" for a computer-estimated starting point.
   - Note: Values are automatically swapped if the selection becomes inverted.

Step 3: Define Goals

1. Set the **Mode** (usually "Imaging" or "Line").
2. Enter your **Washout** time (can be applied from the SPR Tab).
3. Set your target quality metrics (e.g., **Sync Target (Pulses)** = 10, **Target SNR** = 10).
4. Select a **Sync Strategy** to dictate how calculations are prioritized.

Step 4: Run Optimization

The optimization runs automatically whenever you change a parameter.

1. Watch the **Optimised Settings** panel.
2. The software will display the recommended **Spot Size**, **Rep-Rate**, and **Speed**.
3. Check the **Results Table** at the bottom. Are any important isotopes failing to reach the target SNR?
   - *Orange Highlight*: Cannot reach Target SNR (set to Min).
   - *Blue Highlight*: Hardware limited (cannot go faster).
4. If the Spot Size is too large/small, adjust the **Target SNR** or **Sync Target (Pulses)** to constrain it.

Step 5: Advanced Logic & Overrides

1. **Min SNR Constraints**:
   - **Quadrupoles**: If a channel cannot meet the Minimum SNR, its dwell time is forced to the *Minimum Dwell* allowed by hardware to save time for other channels.
   - **MC / TOF**: Since all channels are measured simultaneously, the dwell time cannot be reduced individually. Instead, the channel is flagged with an Orange warning in the table.
2. **Min Duty Cycle**:
   - Forces the total integration time to be long enough relative to the magnet settling time (overhead). Useful for ensuring you aren't wasting 50%+ of your analysis time just switching masses.

3. **Manual Overrides**:
    - If you need to force a specific spot size:
    - Change the **Spot Size** spinbox in the "Optimised Settings" panel (it normally says "Auto").
    - This locks the spot size and re-optimises only the Rep-Rate and Speed.
    - Set it back to 0 (or "Auto") to resume full optimization.
4. If you need to force a specific rep-rate:
    - This is usually derived from the spot size and washout. To influence it, try changing the **Avoid Gaps** setting or **Washout** value.

Tips & Tricks

- **Linking SPR**: Use the SPR Tab first to determine the exact washout time for your cell/tubing, then "Apply" it to the Optimiser.
- **Duty Cycle**: For Sequential ICP-MS (Quadrupoles), a higher Duty Cycle means more efficient use of time. For TOF/MC, this is always 100%.
- **Avoid Gaps**: Essential for imaging. It ensures that `Speed <= Spot Size * Rep Rate`, preventing "striping" artifacts in your map.

Status Messages

The Optimiser provides feedback through the status bar and log to explain why certain settings were chosen or constrained. Understanding these messages can help you troubleshoot why a specific result was calculated.

General Optimization Messages

- **"Optimum Spot Size [X] µm - Based on Minimum SNR of [Isotope]"**
- Indicates that the Spot Size was calculated specifically to satisfy the SNR target for the weakest isotope listed.
- **"Optimum Spot Size [X] µm - Overridden to [Y] µm"**
- Appears when you have manually set the Spot Size spinbox to a specific value (Y), overriding the calculated optimum (X).

Hardware Constraints

- **"Constraint: Acq Time increased for Rep Rate Limit"**
- The Acquisition Time had to be extended because the required Dwell Times + Washout would require a Laser Rep-Rate higher than the laser can physically fire.
- **"Constraint: Acq Time increased for Stage Speed Limit"**
- The Acquisition Time had to be extended because the stage would need to move faster than the cell's maximum speed to cover the spot size in the calculated time.
- **"Constraint: Acq Time increased for Duty Cycle ([X]%)"**

- The total integration time was increased to ensure that the measurement time vs. overhead (settling) time met your "Min Duty Cycle" target.
- **"Constraint: Acq Time increased for Dwell Budget ([X] ms)"**
- The total time was increased because the minimum required dwell times for all elected isotopes exceeded the initial budget derived from the Washout Time. Note: For Simultaneous systems (TOF / Multi-Collector), this budget represents the *maximum single* active dwell assignment required. For Sequential systems (Quadrupole), this budget is the *cumulative sum* of all dwell assignments.

Channel-Specific Warnings

These messages list specific isotopes that are affecting the optimization logic.

- **"The following channels do not meet the minimum SNR target..."**
- **Orange Warning**: These channels have a calculated SNR below your "Minimum SNR" threshold.
- For **MC/TOF** systems, this is just a warning.
- For **Quadrupole** systems, these channels have been forced to the minimum dwell time to save budget.
- **"The following channels cannot be set lower than the hardware minimum ([X] ms)..."**
- **Blue Warning**: The algorithm wants to reduce the dwell time for these channels (usually high-signal ones) to optimise speed, but is blocked by the hard limit of the instrument (e.g., 1ms or 10ms minimum dwell).
- **"The following channels had zero counts in the background. The detection limit (L_c) has been calculated using the Square Root Transform rule..."**
- **Gray Warning**: Indicates that the background signal was perfectly zero (common in simulated data or extremely short baselines). To prevent division-by-zero errors in the SNR calculation, a theoretical detection limit (L_c) was calculated using the Square Root Transform rule for variance stabilization. The results matrix displays these estimated targets with a grey `(Lc)` subscript.

Scientific References

The algorithms and logic used in the iolite Optimiser are based on established statistical principles and laser ablation literature:

- **SNR and Detection Limits**:
- Tanner, S. D. (2010). "Shorter signals for improved signal to noise ratio, the influence of Poisson distribution" *Journal of Analytical Atomic Spectrometry (JAAS)*, 25, 405–407.
- Donard, A., et al. "Determination of relative rare earth element distributions in very small quantities of uranium ore concentrates using femtosecond UV laser ablation– SF-ICP-MS coupling" J. Anal. At. Spectrom., (2015), 30, 2420–2428
- **Optimization Workflow & Loop Logic**:
- Van Malderen, S. J., et al. "Considerations on data acquisition in laser ablation-inductively coupled plasma-mass spectrometry with low-dispersion interfaces" Spectrochimica Acta Part B, (2018), 140, 29–34
- Van Elteren, J. T., et al. "Insights into the selection of 2D LA-ICP-MS (multi) elemental mapping conditions" J. Anal. At. Spectrom., (2019), 34, 1919

- Lockwood, E. T. (2024). "Multiplexed elemental bioimaging with quadrupole ICP-MS andhigh-frequency laser ablation systems" J. Anal. At. Spectrom., 39, 1125
- **Single Pulse Response and Washout Analysis**:
- Ulianov, A., et al. "The ICPMS signal as a Poisson process: a review of basic concepts" J. Anal. At. Spectrom., (2015), 30, 1297–1321
- **Statistical Framework**:
- Currie, L. A. "Limits for qualitative detection and quantitative determination: Application to Radiochemistry." Anal. Chem., (1968), 40, 586-593.
- Currie, L. A. "The Measurement of Environmental Levels of Rare Gas Nuclides and the Treatment of Very Low-Level Counting Data", IEEE Trans. Nucl. Sci., (1972), NS19, (1), 119-126.